ENDOG mtDNA-breakpoint motif-enrichment hypothesis — Deep Research Report

Gene: ENDOG (Q14249), *Homo sapiens* (NCBITaxon:9606) **Hypothesis slug:** `mtdna-breakpoint-motif-enrichment` **Focus type:** free_text (computational analysis on sequence + coordinate data) **Date:** 2026-10-26

> **Seed hypothesis.** Human ENDOG's cleavage-site preferences explain where mitochondrial DNA breakpoints fall: breakpoints of nuclear mitochondrial insertions (NUMTs) **and** of mtDNA deletions are enriched at G-quadruplex-forming and C-rich motifs in the mitochondrial genome, relative to position- and composition-matched control sites.

This was executed as a **computation on sequence and coordinate data**, not a literature review. All breakpoint coordinate sources named in the seed context were obtained programmatically.

Executive Judgment

Verdict: PARTIALLY SUPPORTED, and over-interpreted as stated — the NUMT arm is REFUTED and the deletion arm is a direct-repeat confound rather than evidence of ENDOG causation.

- **mtDNA deletions:** breakpoints *are* statistically enriched at G4-forming / C-rich motifs, even after matching local GC composition (local max|G4Hunter| fold ≈ 1.17, z = +13, p < 1e-4; robust across G4Hunter windows and a stringent ±1 kb local null). **However**, a direct-repeat stratification shows the entire signal comes from **repeat-mediated deletions** (DR ≥ 5 bp: fold 1.24, z = +16) and **disappears in non-repeat deletions** (DR < 5 bp: fold 0.97, p = 0.91). mtDNA direct repeats are themselves G/C-rich, so the "motif" enrichment is **collinear with the classical slipped-replication / direct-repeat mechanism** and does not isolate ENDOG cleavage.

- **NUMTs:** No enrichment at G4/C-rich motifs over GC-matched controls, in **three independent datasets** — the exact 66,083-genome catalogue named in the seed (P 36198798 germline, z = +0.3, p = 0.39; somatic, p = 0.10) and the UCSC NumtS set (p = 0.59). The seed's NUMT claim is **refuted**.

Curation bottom line: This computational result should **not** be used to assign ENDOG a mtDNA-breakpoint-determining function. It demonstrates *co-localization* of deletion breakpoints with G/C-rich, repeat-prone sequence — a long-known property of mtDNA deletions — not a specific, ENDOG-attributable cleavage signature. The enrichment is real but non-discriminating; the NUMT prediction is wrong.

Computed Results (primary provenance)

GC-composition-matched permutation null (10 GC deciles, 10,000 permutations); statistic = local max|G4Hunter| (window 25, ±5 bp slack) and local C-fraction (window 25). Reference: rCRS NC_012920.1 (16,569 bp; reference/light strand is C-rich: C = 31.3 %, G = 13.1 %, so **negative G4Hunter = C-rich on the reference = G4-forming potential on the complementary heavy strand**).

Breakpoint set	n positions	max|G4| obs	null	fold	z	p	C-frac fold	C-frac z	C-frac p
mtDNA deletions (MitoBreak, all)	1963	0.794	0.680	**1.168**	+13.0	1e-4	1.080	+12.3	1e-4
— deletions: direct-repeat DR≥5 bp	1483	0.847	0.685	**1.237**	+16.2	1e-4	1.115	+15.5	1e-4
— deletions: non-repeat DR<5 bp	609	0.647	0.667	**0.969**	−1.4	0.91	0.973	−2.3	0.99
NUMT germline (P 36198798 T1)	2758	0.670	0.669	1.003	+0.2	0.40	1.009	+1.6	0.051
NUMT somatic (P 36198798 T4)	628	0.681	0.662	1.029	+1.3	0.10	1.018	+1.5	0.068
NUMT UCSC NumtS	1195	0.638	0.640	0.997	−0.2	0.59	0.994	−0.7	0.75

Supporting robustness (deletions, all): - **Parameter sweep:** fold 1.14–1.28 across G4Hunter windows 15/25/35 × ±0/5/10 bp slack, all p ≈ 2e-4. - **Stringent local null (±1 kb):** breakpoints sit at the 57th percentile (median 60th) of G4 propensity *within their own neighborhood* (Wilcoxon p = 2.8e-23) → not purely a regional-composition artifact. - **Mann–Whitney vs whole genome:** deletions p = 4.4e-43; NUMTs (germline) p = 0.015 (unmatched, dissolves under GC-matching). - **Known hotspots sit at extreme motifs:** 9 bp deletion hotspot 8271–8281 max|G4| = 2.0 (99.8th genome percentile); CSB2 (99.9th); common 4977-del 5′ 8470–8482 (96.9th), 3′ 13447–13459 (90.3th). These are all direct-repeat-flanked deletions, i.e., the hotspot G/C-richness *is* the

repeat. - **Direct-repeat prevalence:** 69.9 % of the 1,369 deletion pairs have a ≥5 bp perfect repeat at the junction (median 5 bp), consistent with the known microhomology/direct-repeat mechanism.

Data provenance (all fetched programmatically this run): - rCRS: NCBI efetch `NC_012920.1`. - mtDNA deletions: MitoBreak human `Del` table (1,369 pairs → 1,963 unique breakpoints). - NUMTs: Springer static-content supplementary of

P 36198798

(`MOESM4_ESM.xlsx`, Suppl. Tables 1 & 4); UCSC hg19 `numtSMitochondrion` track (766 features → 1,195 boundaries).

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confiden limitatio
This analysis (MitoBreak + rCRS)	Computational	Qualifies	Deletion breakpoints enriched at G4/ C-rich motifs vs matched controls	Fold 1.17, z +13, p<1e-4, GC-matched; robust to params and local null	Human mtDNA deletions, 1,963 positions	High for associatio cannot attribute ENDOG
This analysis (DR stratification)	Computational	Competing/ Refutes causal reading	Is the enrichment independent of direct repeats?	Signal only in DR≥5 bp deletions (fold 1.24); absent in DR<5 bp (fold 0.97, p=0.91)	Human mtDNA deletions	High; LCS proxy for repeats (- bp)
This analysis (P 36198798 T1/T4)	Computational	Refutes	NUMT breakpoints enriched at G4/ C-rich motifs	No enrichment after GC-matching (germline p=0.39; somatic p=0.10)	1,637 germline + 379 somatic NUMTs, 66,083 genomes	High; authorita intended dataset
This analysis (UCSC NumtS)	Computational	Refutes	NUMT breakpoints enriched (replication)	No enrichment (p=0.59)	766 NUMTs	High; independ dataset
P 36198798 (Suppl. Table 6)	Database/ computational (their own)	Qualifies	Breakpoint trinucleotide motif enrichment	Mixed: C-rich (CCC p=2e-5, ACC p=4e-7) *and* AT-rich (TTT p=7e-5, ATT p=4e-5)	NUMT mtDNA/ nuclear breakpoints	Microhor signal, no clean G4/
P 36394256	Direct assay (cited background)	Supports (motif biology)	ENDOG binds G4 at 9 bp hotspot, induces breaks	Provides the mechanistic rationale for	Human, in vitro/cell	Not re-de here; mo

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confiden limitatio
				motif relevance		biology ta as given
P 39752519	Direct assay (cited background)	Supports (motif biology)	ENDOG cleaves C-rich motifs	Mouse mitochondrial transcript, C-rich	Mouse	Organism substrate specific (context)
P 41854021	Direct assay (cited background)	Qualifies	ENDOG prefers oxidatively-damaged/ nicked/gapped DNA	Primary determinant may be lesion/ structure, not plain sequence motif	Human, in vitro	Suggests damage, motif, dri site selec
P 39227600	Mutant phenotype (cited background)	Competing	Yeast EndoG limits long-mtDNA→nucleus transfer	If conserved, predicts ENDOG shapes NUMT ends — not supported by human NUMT coords	*S. cerevisiae*	Organism specific; human N test is nu

GO Curation Implications (leads — require curator verification)

- **Do NOT create/assign a BP term implying ENDOG determines mtDNA breakpoint/ NUMT positions** (e.g., nothing like "regulation of mitochondrial genome rearrangement / NUMT formation via sequence-specific cleavage"). The computational enrichment does not support an ENDOG-specific, motif-driven breakpoint-placement function; the deletion signal is a direct-repeat confound and the NUMT prediction is refuted.

- **Molecular function:** Evidence (cited, not re-derived) supports ENDOG's core MF as a **nuclease acting on structured/damaged DNA** — candidates such as `endonuclease activity` (GO:0004519) / `deoxyribonuclease activity` (GO:0004536), and possibly `G-quadruplex DNA binding` (GO:0051880) on the strength of

  P 36394256

  These are MF claims about the *enzyme*, independent of the breakpoint-enrichment hypothesis, and should be curated from the primary assays, not from this coordinate analysis.
- **This analysis is best logged as negative/qualifying computational context**, not as an IEA/ISS basis for any term. Prefer: retain existing experimentally-supported nuclease MF terms; **do not** add a breakpoint-placement BP term on the basis of motif co-localization.
- Avoid "protein binding" as a recommendation; a more informative supported MF is the nuclease / G4-binding activity above (from primary literature, pending curator check of those PMIDs).

Mechanistic Scope

- **Immediate molecular activity under test:** sequence/structure-specific DNA cleavage by ENDOG (G4-forming / C-rich motif preference). The analysis probes whether *where breakpoints sit in the mtDNA sequence* matches ENDOG's reported preference.
- **What the data actually show:** a statistical **association** between mtDNA *deletion* breakpoints and G/C-rich, G4-prone, **direct-repeat** sequence. This is a property of the breakpoints' genomic neighborhood, not a demonstration that ENDOG cut there.
- **Downstream / not tested here:** actual deletion/NUMT formation, replication-slippage vs nuclease-cut causation, ENDOG loss-of-function effects on breakpoint spectra, disease phenotypes. The enrichment is agnostic to which nuclease (if any) acts; direct-repeat replication slippage explains the same localization without invoking ENDOG.

Conflicts and Alternatives

1. **Direct-repeat mechanism (strongest competing explanation).** 70 % of deletions are flanked by ≥5 bp repeats; the motif enrichment is entirely within this class. mtDNA deletion hotspots have long been attributed to direct/imperfect repeats and replication slippage — a mechanism that does not require ENDOG and fully accounts for the observed G/C-rich localization.
2. **NUMT arm contradicted by its own source.** The 66,083-genome catalogue shows no G4/C-rich enrichment after composition matching; its trinucleotide table is dominated by low-complexity microhomology (AT- and C-rich), consistent with non-homologous end joining at the nuclear insertion site rather than ENDOG motif cleavage in mtDNA.
3. **Substrate determinant may be damage, not sequence** **(** P 41854021 **).** If ENDOG's site selection is driven by oxidative lesions / nicks / gaps, a static reference-sequence motif scan cannot capture it — the negative/confounded motif result is expected even if ENDOG is biologically involved.
4. **Organism / substrate specificity.** The C-rich cleavage evidence is mouse (P 39752519) and in an RNA/transcript context; the yeast transfer-limiting role (P 39227600) is species-specific. Neither licenses a human DNA-breakpoint-placement claim.
5. **Composition collinearity.** G4 propensity, C-richness, and GC content are mutually correlated in the GC-skewed major arc where deletions cluster; GC-decile matching mitigates but cannot fully disentangle these from repeat content.

Knowledge Gaps

- **Causal attribution to ENDOG.** Checked: motif co-localization (association only). Why it matters: association ≠ cleavage. Resolve with ENDOG-knockout vs WT mtDNA breakpoint spectra (ChIP/cleavage-seq, END-seq, or deletion mapping).

- **Damage-conditioned motif preference.** Checked: static rCRS scan only. Why it matters: ENDOG prefers oxidized/nicked DNA (P 41854021), invisible to reference-sequence scans. Resolve with in-vitro cleavage mapping on damaged mtDNA templates, then test breakpoint overlap.
- **Repeat-independent deletions.** Checked: DR<5 bp subset is null. Why it matters: this is the cleanest window to detect an ENDOG-specific motif effect, and it shows none. Resolve with a larger, uniformly-ascertained repeat-free deletion set.
- **G4 strand/structure validation.** Checked: G4Hunter propensity (sequence predictor). Why it matters: predicted ≠ folded G4. Resolve with experimental G4 maps (BG4 CUT&Tag, G4-seq) on mtDNA and overlap with breakpoints.

Discriminating Tests

1. **ENDOG-null vs WT breakpoint spectra** (cells/mouse): if ENDOG drives breakpoint placement, its loss should shift/deplete breakpoints at G4/C-rich sites — most informative single experiment.
2. **In-vitro ENDOG cleavage-site mapping** on mtDNA (undamaged vs oxidatively-damaged) → define the true human cleavage motif, then test breakpoint overlap (replaces the assumed motif with a measured one).
3. **Repeat-controlled re-analysis** at scale: restrict to repeat-free deletions and NUMTs and ask whether *any* residual ENDOG-specific motif survives (this run: none).
4. **Experimental G4 maps (G4-seq / BG4)** on mtDNA to replace G4Hunter predictions and re-test overlap.
5. **Cross-nuclease comparison:** test whether other mitochondrial nucleases' known motifs fit the breakpoints equally well — if so, the signal is non-specific.

Curation Leads (require curator verification)

- **Action lead:** Treat the breakpoint-motif hypothesis as **non-core / not curation-actionable for a breakpoint-placement function**. Flag the deletion enrichment as **confounded by direct repeats** and the NUMT enrichment as **refuted** in the review free-text.
- **Candidate references to verify (exact snippets to pull from primary sources):**
- P 36394256 — verify the specific claim that ENDOG **binds G-quadruplex** at the 9 bp hotspot and induces breaks (supports a `G-quadruplex DNA binding` MF, GO:0051880, if the assay is direct).
- P 41854021 — verify ENDOG's preference for **oxidatively-damaged / nicked / gapped DNA** (argues site selection is lesion-driven, tempering any pure-motif claim).
- P 36198798 — the authoritative NUMT coordinates used here; Suppl. Table 6 trinucleotide result is mixed, not G4/C-rich-specific (supports the refutation).
- **Candidate GO terms (MF, from primary assays — not from this analysis):** `endonuclease activity` (GO:0004519) / `deoxyribonuclease activity` (GO:0004536); conditionally `G-quadruplex DNA binding` (GO:0051880). **Do not** add a breakpoint-/NUMT-placement BP term.
- **Suggested curator questions:** (i) Is there any *ENDOG-dependent* (knockout-controlled) evidence that it determines human mtDNA breakpoint positions? (ii) Does the review already cite the direct-repeat mechanism as the explanation for deletion hotspots? (iii) Is the G4-binding claim (P 36394256) a direct binding/cleavage assay suitable for an MF annotation?
- **Suggested experiment:** ENDOG-null vs WT mtDNA breakpoint mapping (above) as the decisive test.

Limitations

- Reference-sequence motif scan cannot capture damage-conditioned or structure-conditioned cleavage.
- Direct-repeat detection used a longest-common-substring proxy (±15 bp); imperfect repeats and microhomology thresholds could shift the exact DR≥5 split, though the qualitative confound is robust.
- G4Hunter predicts, but does not prove, G4 folding.
- MitoBreak aggregates deletions of heterogeneous ascertainment; recurrent hotspots were collapsed to unique positions to avoid double-counting (results unchanged).
- No ENDOG perturbation data were available; causation cannot be tested from coordinates alone.

Conclusion

The seed hypothesis is **partially supported for the descriptive claim** (deletion breakpoints co-localize with G4/C-rich motifs) but **over-interpreted as an ENDOG function**: the deletion signal is a **direct-repeat confound** (absent in non-repeat deletions), and the **NUMT arm is refuted** in the exact 66,083-genome catalogue and a second independent dataset. The analysis should inform the review as a **negative/qualifying computational result**, not as support for a new ENDOG breakpoint-placement annotation.